complaint is subjective at best, and the latter conforms to the nonlocality Bell proved unavoidable, so neither criticism is convincing. Nevertheless, perhaps unjustifiably, Bohm's approach has never caught on.[13] The Ghirardi-Rimini-Weber approach deals with wavefunction collapse directly, by changing the equations to incorporate a new spontaneous collapse mechanism. But, detractors point out, there is as yet not a shred of experimental evidence supporting the proposed modification to Schrödinger's equation.

Research seeking a solid and fully transparent connection between the formalism of quantum mechanics and the experience of everyday life will no doubt go on for some time to come, and it's hard to say which, if any, of the known approaches will ultimately achieve a majority consensus. Were physicists to be polled today, I don't think there would be an overwhelming favorite. Unfortunately, experimental input is of limited help. While the Ghirardi-Rimini-Weber proposal does make predictions that can, in certain situations, differ from standard stage one / stage two quantum mechanics, the deviations are too small to be tested with today's technology. The situation with the other three proposals is worse because they stymie experimental adjudication even more definitively. They agree fully with the standard approach, and so each yields the same predictions for things that can be observed and measured. They differ only regarding what happens backstage, as it were. They only differ, that is, regarding what quantum mechanics implies for the underlying nature of reality.

Even though the quantum measurement problem remains unsolved, during the last few decades a framework has been under development that, while still incomplete, has widespread support as a likely ingredient of any viable solution. It's called *decoherence.*

Decoherence and Quantum Reality

When you first encounter the probabilistic aspect of quantum mechanics, a natural reaction is to think that it is no more exotic than the probabilities that arise in coin tosses or roulette wheels. But when you learn about quantum interference, you realize that probability enters quantum mechanics in a far more fundamental way. In everyday